

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
import numpy as np

def bipolar_sigmoid(net):
    return (2 / (1 + np.exp(-net))) - 1

def bipolar_sigmoid_derivative(output):
    return 0.5 * (1 - output**2)

weights = np.array([1, -1, 0, 0.5])
learning_rate = 0.1

input_vectors = [
    np.array([1, -2, 0, -1]),
    np.array([0, 1.5, -0.5, -1]),
    np.array([-1, 1, 0.5, -1])
]

desired_outputs = [-1, -1, 1]

for epoch in range(2):
    print(f"Epoch {epoch + 1}:")
    for i, x in enumerate(input_vectors):
        net = np.dot(weights, x)
        output = bipolar_sigmoid(net)
        error = (desired_outputs[i] - output)
        derivative = bipolar_sigmoid_derivative(output)
        delta_w = learning_rate * error * derivative * x
        weights += delta_w

    print(f" Input vector x{i + 1}: {x}")
    print(f" Net{i + 1}: {net}")
    print(f" Output{i + 1}: {output:.4f}")
    print(f" Derivative{i + 1}: {derivative:.4f}")
    print(f" Updated Weights: {weights}\n")
```

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

Output:

```
Epoch 1:
Input vector x1: [ 1 -2  0 -1]
Net1: 2.5
Output1: 0.8483
Derivative1: 0.1402
Updated Weights: [ 0.97408569 -0.94817138  0.          0.52591431]

Input vector x2: [ 0.   1.5 -0.5 -1. ]
Net2: -1.948171379043783
Output2: -0.7505
Derivative2: 0.2184
Updated Weights: [ 0.97408569 -0.95634441  0.00272434  0.531363  ]

Input vector x3: [-1.   1.   0.5 -1. ]
Net3: -2.46043092567365
Output3: -0.8426
Derivative3: 0.1450
Updated Weights: [ 0.94737155 -0.92963027  0.01608141  0.50464886]
```

```
Epoch 2:
Input vector x1: [ 1 -2  0 -1]
Net1: 2.301983232259518
Output1: 0.8181
Derivative1: 0.1654
Updated Weights: [ 0.9173058  -0.86949877  0.01608141  0.53471461]

Input vector x2: [ 0.   1.5 -0.5 -1. ]
Net2: -1.8470034741168142
Output2: -0.7275
Derivative2: 0.2353
Updated Weights: [ 0.9173058  -0.87911636  0.01928728  0.54112633]

Input vector x3: [-1.   1.   0.5 -1. ]
Net3: -2.3279048562681854
Output3: -0.8223
Derivative3: 0.1619
Updated Weights: [ 0.88780387 -0.84961443  0.03403824  0.5116244  ]
```